Algebra 1: Polynomials

Foundation Level

Name: ____________________
Date: ____________________

Question 1: What is the degree of the polynomial $3x^2 + 2x - 1$?

A) 1
B) 2
C) 3
D) 4
E) 5

Question 2: Simplify the expression $2x^2 + 5x + 3 + 2x^2 - 3x - 2$ to get?

A) $3x^2 + 2x + 1$
B) $4x^2 + 2x + 1$
C) $4x^2 + 8x + 1$
D) $4x^2 + 2x + 0$
E) $4x^2 + 2x + 2$

Question 3: If $x = 2$, what is the value of $x^2 + 3x$?

A) 4
B) 6
C) 7
D) 10
E) 12

Question 4: Factor the polynomial $x^2 + 5x + 6$ into?

A) $(x + 2)(x + 3)$
B) $(x + 1)(x + 6)$
C) $(x - 2)(x - 3)$
D) $(x - 1)(x - 6)$
E) $(x + 2)(x - 4)$

Question 5: What is the value of $(2x^2 + 1) - (x^2 + 2x)$ when $x = 1$?

A) -1
B) 0
C) 1
D) 2
E) 3

Question 6: Given the polynomial $x^3 - 6x^2 + 11x - 6$, what is its factored form?

A) $(x-1)(x-2)(x-3)$
B) $(x-1)(x+2)(x+3)$
C) $(x+1)(x-2)(x-3)$
D) $(x-1)(x+1)(x-6)$
E) $(x-2)(x-2)(x-3)$

Question 7: If $f(x) = x^2 - 4$, what is $f(0)$?

A) -4
B) -2
C) 0
D) 2
E) 4

Question 8: Simplify $(3x^2 + 2) + (2x^2 - 3x)$ to get?

A) $5x^2 - 3x - 1$
B) $5x^2 - 3x + 2$
C) $5x^2 + 3x + 2$
D) $5x^2 - 3x + 0$
E) $5x^2 + 3x - 1$

Question 9: Solve for x in the equation $x^2 + 4x + 3 = 0$ and provide the steps to get the solutions.

Question 10: Given $f(x) = 2x^2 + x - 1$, find $f(-2)$ and show all steps to simplify the expression.

Chef's Tips (Hints):

- Pay close attention to the signs and the order of operations when simplifying polynomial expressions.
- For factoring polynomials, look for two numbers that multiply to the constant term and add up to the coefficient of the x term.
- When solving quadratic equations by factoring, set each factor equal to zero and solve for x.